Passive Acoustic Sensor Concept Demo: Executive Summary

Nov 10, 2026 · @Shane

Bottom line

This concept demo turned an early software prototype of a passive acoustic drone sensor into a complete, checked design package. The package exists in software and on paper. **No hardware has been bought, built, or tested.**

The package is release 0.2.2, dated 10 November 2026. It holds the requirements, a priced bill of materials, engineering drawings, a firmware setup procedure, working software, test procedures, and an interactive demo. Leadership can open the demo and the drawings today without installing anything.

The sensor is passive. It listens with four microphones, flags rotor-like sound, and estimates the direction the sound comes from. It does not identify aircraft, measure range, or transmit anything.

The next decision is whether to fund the physical build. One sensor node needs about $209 in parts, and the test equipment adds about $172, at prices seen on 9 November 2026.

On staffing, the demo supports fewer desk hours in drafting, documentation, and first-pass engineering. It does not yet support eliminating positions, because review, sign-off, and all physical work still need people.

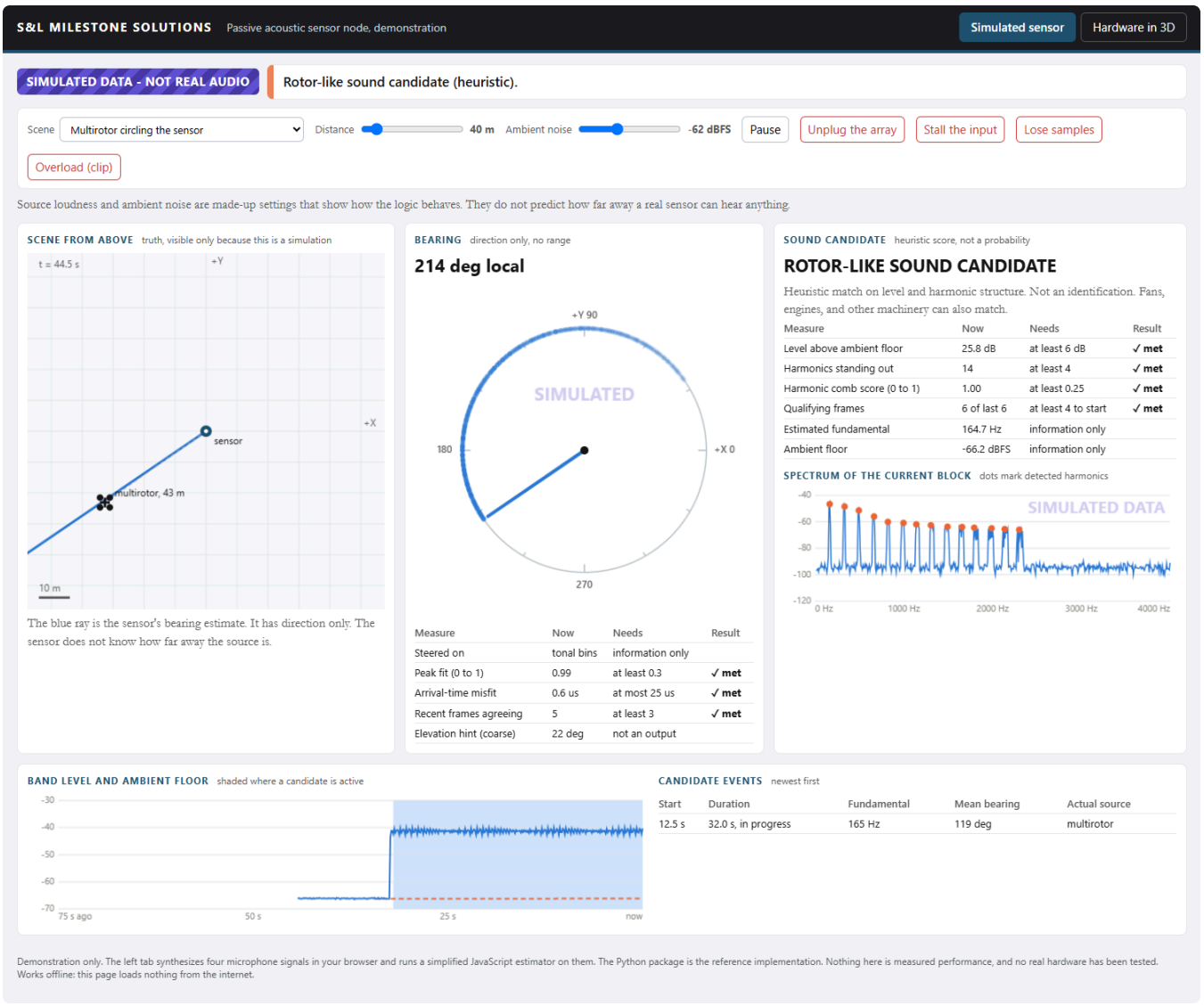
What was created

The exercise produced sixteen deliverables that together describe the product, how to build it, and how to prove it. Each row gives its state today.

Deliverable	What it is	Status today
Baseline audit	Review of the starting prototype (Revision A): what worked, what was missing, and a prioritized issue list	Complete

Deliverable	What it is	Status today
Requirements	64 requirements, each with an acceptance criterion, a verification method, and a status	Complete. Hardware and field items are marked pending
Bill of materials	Priced parts list for one sensor node, optional parts, and test equipment, with part numbers and purchase links	Complete. Nothing has been ordered
Hardware build guide	Wiring, power, mechanical design, tools, and assembly steps with inspection points	Written. Not yet followed on real parts
Engineering drawings	Four ANSI B sheets under the S&L Milestone Solutions title block: carrier plate, foam parts, assembly, and wiring	Revision A, PRELIMINARY. Sign-off names are blank
CAD files	The carrier plate as a DXF cutting file and an OpenSCAD 3D model source	Generated. OpenSCAD file not compiled
Firmware setup procedure	Steps to load, read back, and record the manufacturer's firmware on the microphone array	Written from manufacturer documents. Not yet run on a unit
Sensor software	A Windows program of about 4,600 lines that captures audio, flags rotor-like sound, estimates bearing, and monitors its own health	Works on simulated and injected input. Not run on real audio
Operator dashboard	A local web page that shows the candidate, the bearing, the evidence, and any fault	Checked in a browser in simulated mode
Multi-node tools	An offline position estimate from two or more calibrated nodes, with stated uncertainty	Tested on simulated arrays only

Deliverable	What it is	Status today
Classifier workflow	Tools and a plan to collect, label, train, and evaluate a future sound classifier	Tested on synthetic data. No real dataset and no trained model exist
Test procedures	Nine bench procedures and five field procedures, each with pass criteria	Written. None has been run
Record templates	Six templates for unit, firmware, calibration, evaluation, dataset, and field-session records	Ready to use
Interactive demo	A web page with a simulated sensor and a 3D mock-up of the hardware	Published. Everything in it is simulated or drawn
Verification report and limitations	A plain account of what was tested, what passed, and what remains unproven	Complete for release 0.2.2
Release package	One archive of everything above, with a checksum for every file	Release 0.2.2 built and verified

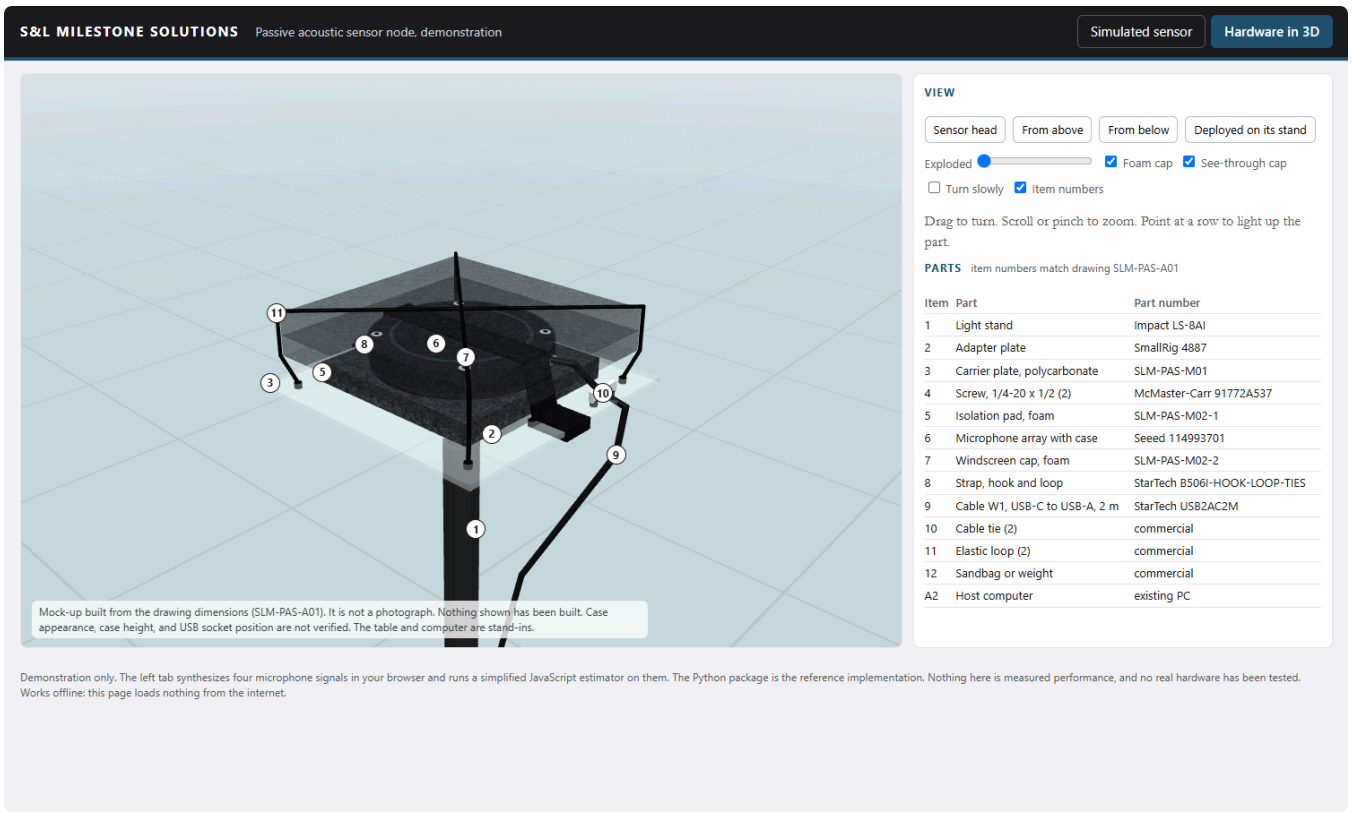


The interactive demo, simulated sensor tab. All data shown is simulated, not real audio.

The hardware and its parts

One sensor node is a stock USB microphone array on a simple mount. It uses commercial parts, one cut plate, and two cut foam pieces. It has no custom electronics.

The array has four microphones in a 66 mm square. It sits on a foam pad on a clear polycarbonate carrier plate, under a foam wind cap, and a strap holds it down. The plate bolts to a tripod adapter on a light stand. One USB cable carries power and audio to an existing Windows 11 PC.



3D mock-up of the sensor head, drawn from the drawing dimensions. It is not a photograph, and nothing shown has been built.

Part	Supplier and part number	Price (US$)
USB four-microphone array with case	Seeed Studio 114993701	59.90
USB cable, USB-C to USB-A, 2 m	StarTech USB2AC2M	20.49
Clear polycarbonate sheet for the carrier plate	McMaster-Carr 8574K26	10.42
Tripod adapter plate	SmallRig 4887	12.99
Screws, 1/4 in-20, pack of 50	McMaster-Carr 91772A537	14.80
Light stand	Impact LS-8AI	32.95
Hook-and-loop ties, pack of 50	StarTech B506I-HOOK-LOOP-TIES	19.46
Foam sheet, 1 in, for the wind cap	The Foam Factory filter foam	24.99
Foam sheet, 1/2 in, for the isolation pad	The Foam Factory filter foam	12.99

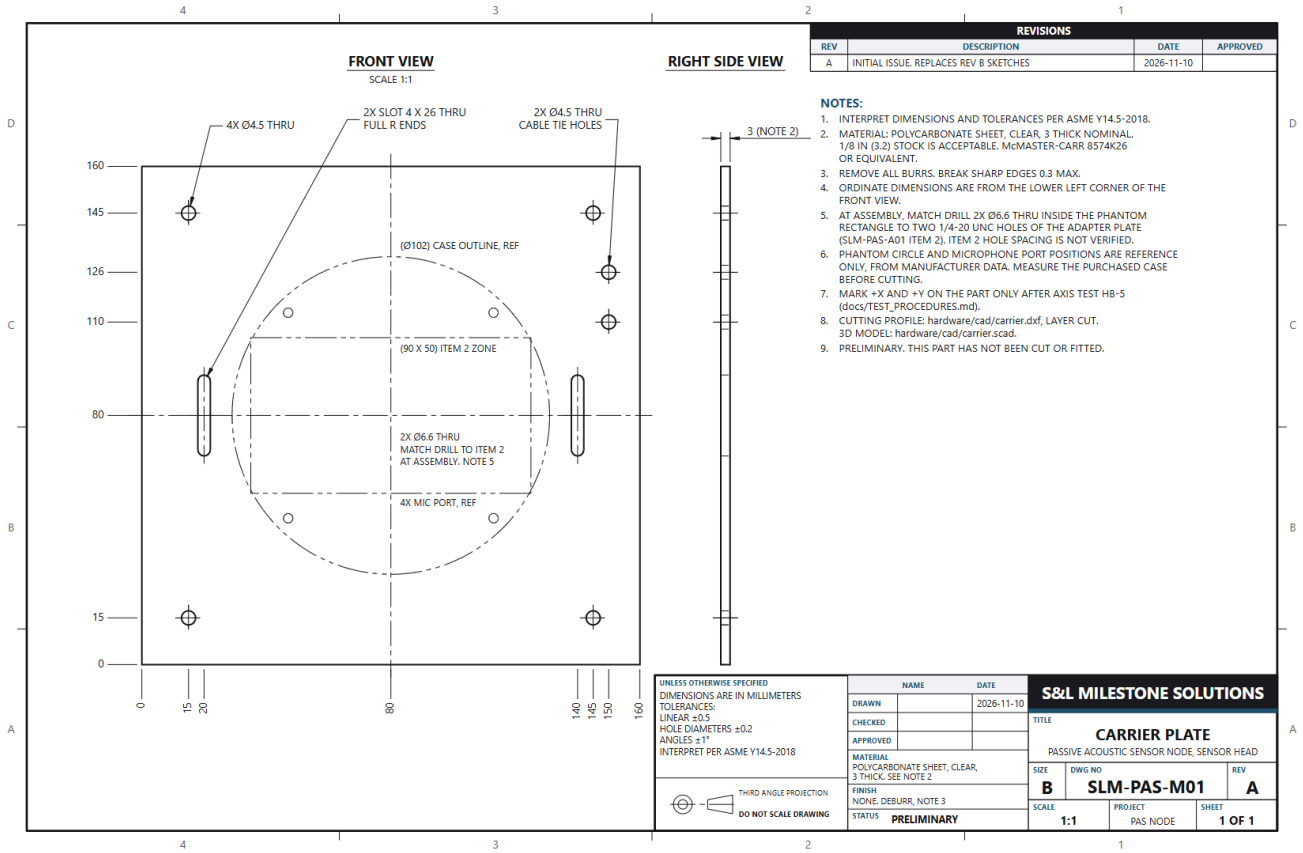
Part	Supplier and part number	Price (US$)
Total for one node		**208.99**

Prices are before tax and shipping, as seen on 9 November 2026. The microphone array was on back order at the supplier's US warehouse that day. Check every price and stock level again before ordering.

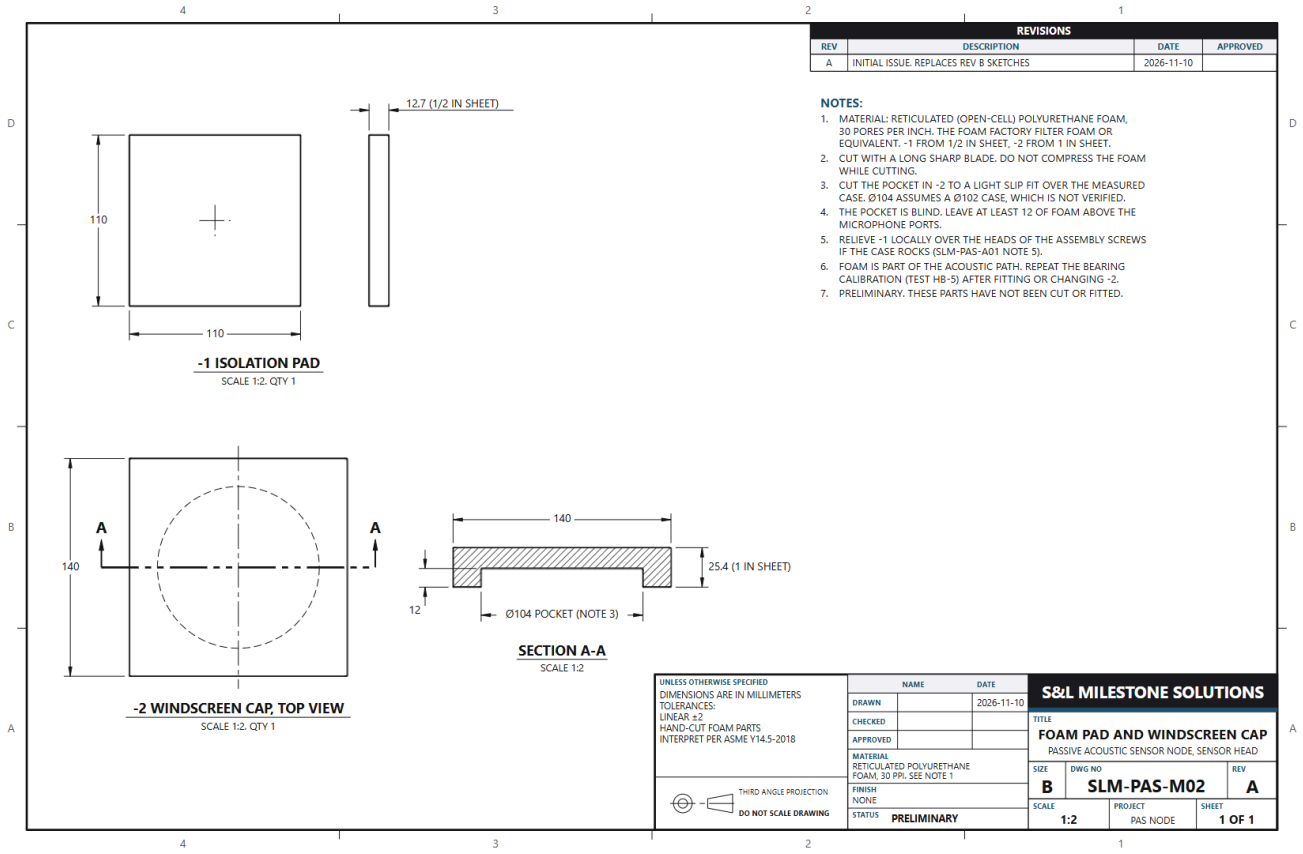
Test equipment adds $171.60: a battery loudspeaker, a tape measure, a thermometer, and an optional sound level meter. A weight for the stand base is required outdoors and is not priced.

The engineering drawings

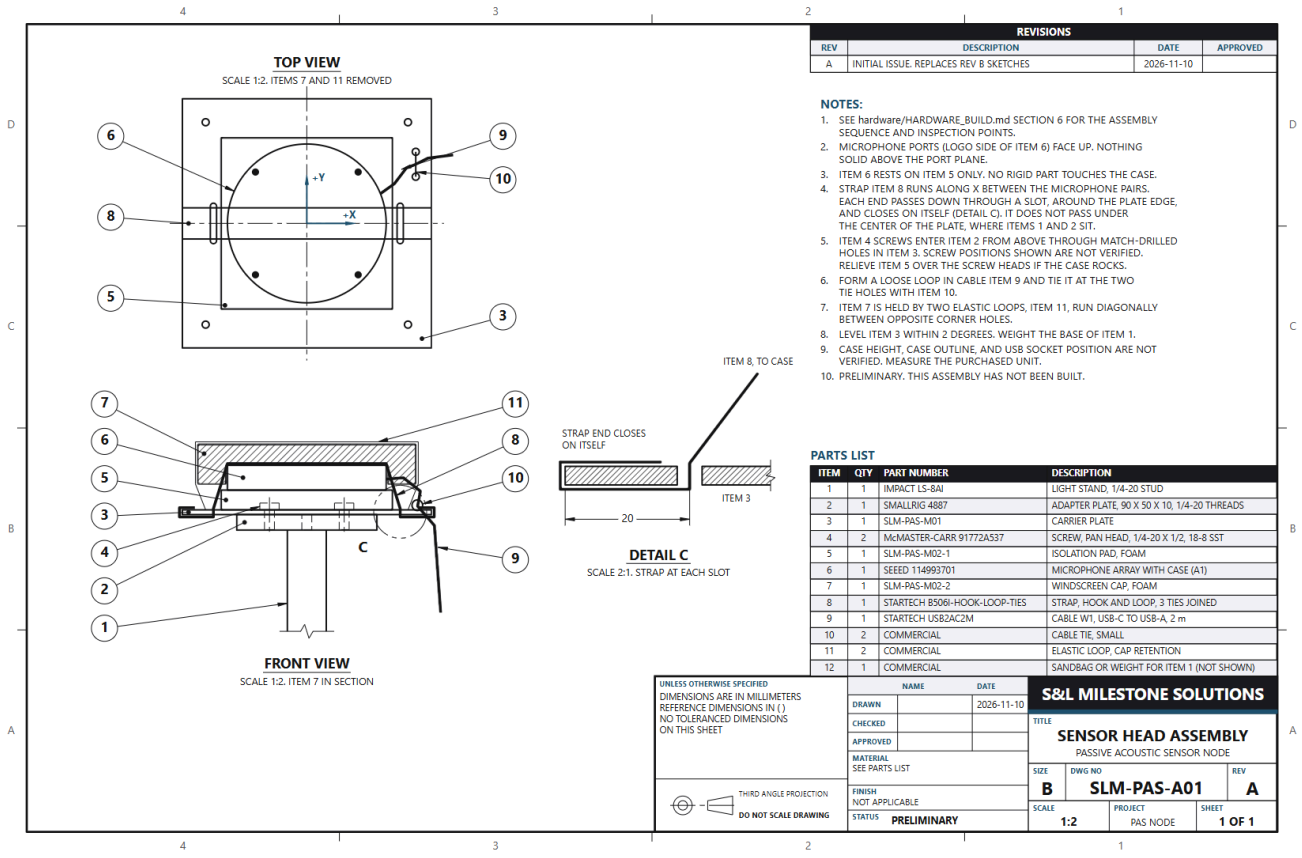
The drawing set has four ANSI B sheets under the S&L Milestone Solutions title block. All four are revision A with status PRELIMINARY, and the DRAWN, CHECKED, and APPROVED names are blank. Nothing has been made from them.



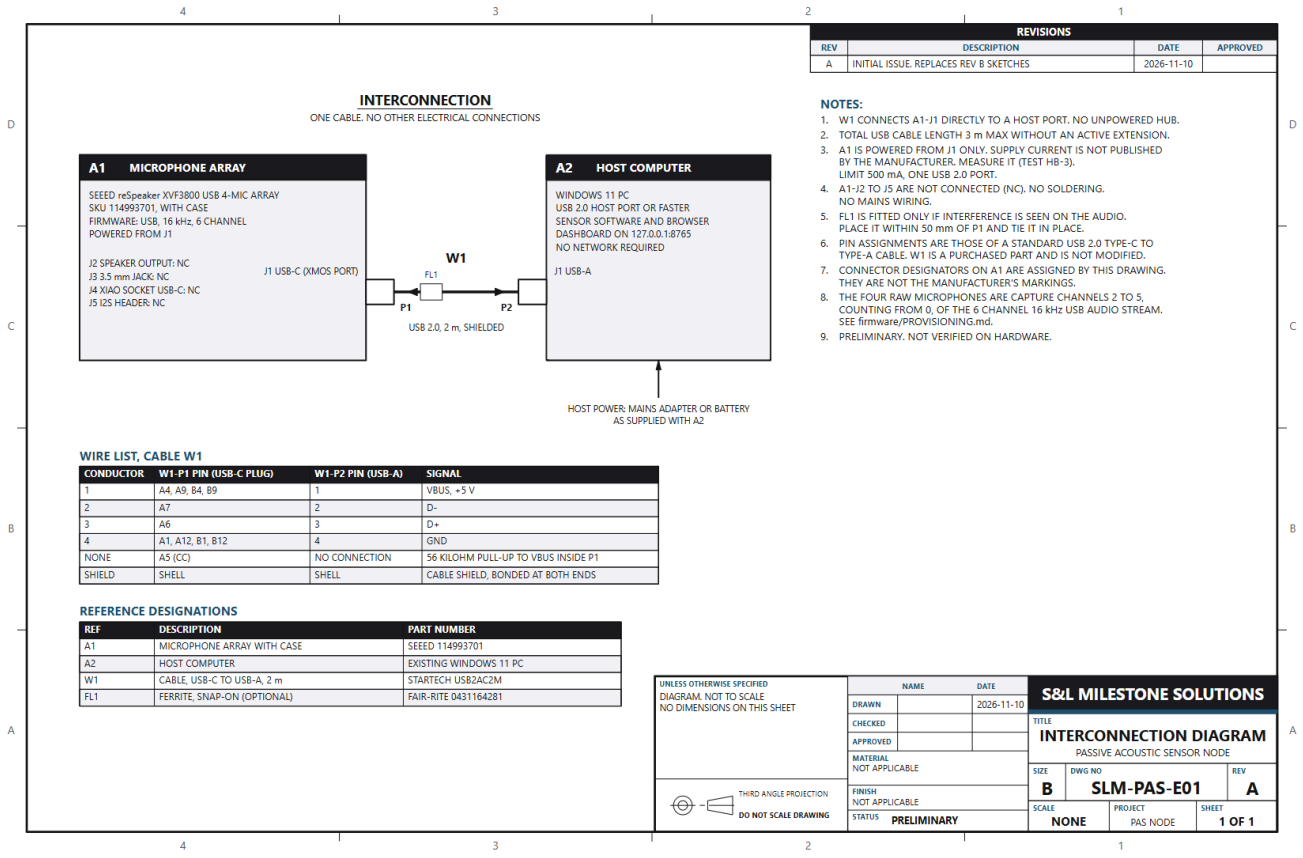
SLM-PAS-M01, carrier plate. This is the one cut part, with hole positions, tolerances, and notes.



SLM-PAS-M02, foam isolation pad and windscreen cap, with section A-A through the cap.



SLM-PAS-A01, sensor head assembly, with numbered parts, the strap detail, and the parts list.



REVISIONS

REV	DESCRIPTION	DATE	APPROVED
A	INITIAL ISSUE. REPLACES REV B SKETCHES	2026-11-10	

INTERCONNECTION

ONE CABLE. NO OTHER ELECTRICAL CONNECTIONS

A1 MICROPHONE ARRAY

SEEED reSpeaker XVF3800 USB 4-MIC ARRAY
SKU 114993701, WITH CASE
FIRMWARE: USB, 16 kHz, 6 CHANNEL
POWERED FROM J1

J2 SPEAKER OUTPUT: NC
J3 3.5 mm JACK: NC
J4 XIAO SOCKET USB-C: NC
J5 I2S HEADER: NC

J1 USB-C (XMOS PORT)

W1
FL1
P1
P2
USB 2.0, 2 m, SHIELDED

A2 HOST COMPUTER

WINDOWS 11 PC
USB 2.0 HOST PORT OR FASTER
SENSOR SOFTWARE AND BROWSER DASHBOARD ON 127.0.0.1:8765
NO NETWORK REQUIRED

J1 USB-A

HOST POWER: MAINS ADAPTER OR BATTERY AS SUPPLIED WITH A2

NOTES:

1. W1 CONNECTS A1-J1 DIRECTLY TO A HOST PORT. NO UNPOWERED HUB.
2. TOTAL USB CABLE LENGTH 3 m MAX WITHOUT AN ACTIVE EXTENSION.
3. A1 IS POWERED FROM J1 ONLY. SUPPLY CURRENT IS NOT PUBLISHED BY THE MANUFACTURER. MEASURE IT (TEST HB-3). LIMIT 500 mA, ONE USB 2.0 PORT.
4. A1-J2 TO J5 ARE NOT CONNECTED (NC). NO SOLDERING. NO MAINS WIRING.
5. FL1 IS FITTED ONLY IF INTERFERENCE IS SEEN ON THE AUDIO. PLACE IT WITHIN 50 mm OF P1 AND TIE IT IN PLACE.
6. PIN ASSIGNMENTS ARE THOSE OF A STANDARD USB 2.0 TYPE-C TO TYPE-A CABLE. W1 IS A PURCHASED PART AND IS NOT MODIFIED.
7. CONNECTOR DESIGNATORS ON A1 ARE ASSIGNED BY THIS DRAWING. THEY ARE NOT THE MANUFACTURER'S MARKINGS.
8. THE FOUR RAW MICROPHONES ARE CAPTURE CHANNELS 2 TO 5, COUNTING FROM 0, OF THE 6 CHANNEL 16 kHz USB AUDIO STREAM. SEE firmware/PROVISIONING.md.
9. PRELIMINARY. NOT VERIFIED ON HARDWARE.

WIRE LIST, CABLE W1

CONDUCTOR	W1-P1 PIN (USB-C PLUG)	W1-P2 PIN (USB-A)	SIGNAL
1	A4, A9, B4, B9	1	VBUS, +5 V
2	A7	2	D-
3	A6	3	D+
4	A1, A12, B1, B12	4	GND
NONE	A5 (CC)	NO CONNECTION	56 KILOHM PULL-UP TO VBUS INSIDE P1
SHIELD	SHELL	SHELL	CABLE SHIELD, BONDED AT BOTH ENDS

REFERENCE DESIGNATIONS

REF	DESCRIPTION	PART NUMBER
A1	MICROPHONE ARRAY WITH CASE	SEEED 114993701
A2	HOST COMPUTER	EXISTING WINDOWS 11 PC
W1	CABLE, USB-C TO USB-A, 2 m	STARTECH USB2AC2M
FL1	FERRITE, SNAP-ON (OPTIONAL)	FAIR-RITE 0431164281

UNLESS OTHERWISE SPECIFIED
DIAGRAM, NOT TO SCALE
NO DIMENSIONS ON THIS SHEET

THIRD ANGLE PROJECTION
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN		2026-11-10
CHECKED		
APPROVED		

MATERIAL: NOT APPLICABLE
FINISH: NOT APPLICABLE
STATUS: PRELIMINARY

S&L MILESTONE SOLUTIONS

TITLE: INTERCONNECTION DIAGRAM
PASSIVE ACOUSTIC SENSOR NODE

SIZE	DWG NO	REV
B	SLM-PAS-E01	A

SCALE	PROJECT	SHEET
NONE	PAS NODE	1 OF 1

SLM-PAS-E01, interconnection diagram and wire list for the single USB cable.

The full-size sheets are in `hardware/drawings/SLM-PAS-DRAWING-SET.pdf`.

What was checked and what passed

Every check so far is software or paper evidence, and release 0.2.2 passes all of them. None of it shows that a built sensor works.

Check	Result
Starting prototype	Its 34 tests passed. Its dashboard file was missing from the package
Automated software tests	178 tests. 177 passed before packaging. The last one checks file checksums and passed after packaging
Fault handling	Injected faults were each reported and recovered from: unplugged device, stalled input, lost samples, clipping, and a dead channel

Check	Result
Operator dashboard	Candidate, stale input, and lost contact displayed correctly in a browser, after one fix
Drawing set	Automated checks found no text collisions, correct title blocks, valid PDF files, and clear plate features. All four sheets were also read by eye
Interactive demo	11 of 11 checks passed in Microsoft Edge, including a phone-width layout check. The published web copy passed its self-test before it went up
Release package	The archive was extracted to a clean folder. All 103 files matched their checksums

The software was also characterized on simulated sound. Those figures describe software behaviour only, and the verification report holds them.

One real measurement exists: on the development PC, the software used about 8 percent of the time available for each audio block.

What is not proven

Nothing about the physical product is proven. No part has been bought, cut, assembled, or connected, and no real sound has been processed.

- **Detection performance.** No detection range, bearing accuracy, false alert rate, or recall has been measured on real sound. The package claims none.
- **Audio capture on Windows.** The software needs six audio channels from the array. Whether Windows provides all six is the first bench test.
- **What the detector can tell apart.** The detector is a heuristic for harmonic sound. Fans, engines, and generators also trigger it. It does not identify drones.
- **Direction only.** One node gives a bearing, not a range or a position. Reflections can produce a wrong bearing that looks right.
- **Drawings.** The drawings are preliminary and unsigned. They are not released for fabrication. Some dimensions come from supplier listings, not from parts in hand.
- **Firmware.** The setup procedure comes from manufacturer documents and has not been run on a unit. The package includes no firmware files.

- **Demo.** The demo uses invented sound levels and a simplified estimator. Its 3D view is a drawing of the design, not a photograph of a built unit.
- **Weather and endurance.** No test covers wind, rain, temperature, or continuous running beyond a few minutes. The product has no environmental rating.
- **Independent review.** An AI assistant produced the package under the owner's direction. No independent engineering review has taken place yet.

What the exercise shows

A hardware product concept can reach a reviewable, costed, and testable state before any money goes to parts. The package's documents are dated 9 and 10 November 2026.

- **Faults were caught on paper.** Drawing the assembly to scale exposed two design faults. The cable-tie holes sat under the strap, and the strap route ran through the mounting plate. Both were corrected before any part was cut.
- **One source of dimensions.** One generator writes the drawings and the cutting file from one set of dimensions. An automated test checks that the 3D mock-up uses the same numbers.
- **Every claim carries a status.** Each of the 64 requirements carries a status, such as verified in software, waiting for a bench test, waiting for a field test, or target only.
- **Anyone can repeat the work.** The tests rerun with one command, the drawings regenerate from code, and checksums verify the release.
- **Problems are on the record.** The verification report lists 12 defects found and fixed along the way, and the limitations document lists what remains open.

What this means for staffing

This demo supports cutting the hours people spend on desk work. It does not yet support eliminating any position, because nothing here was built, signed, or checked by a second person.

Claude (model Fable 5.2) did the first-draft work in this exercise, and the owner directed it. The table shows what that means for each role, based only on what this demo showed.

Role	What Claude did here	What still needs a person	What the demo supports
Drafter	Drew four sheets and a cutting file from code, and checked them by rule	Checking and signing. Fit against real parts. Tolerances and shop feedback. Work in the company's CAD system	Fewer drafting hours on simple parts and assemblies. Keep the checker
Entry-level engineer	Parts research, the bill of materials, first-pass design, test scripts, and document drafts	Bench and field work. Noticing what a draft missed. Growing into the senior reviewer	Fewer junior hours at the desk. Move junior time to the bench and to review
Software developer	Wrote about 4,600 lines of software, 178 tests, and a dashboard	Code review by someone who owns the result. Integration with real hardware. A security review before any network use	One developer who reviews can cover more ground
Technical writer	Wrote every document in the package	A subject owner who confirms each statement is true	Much less drafting time. Review stays
Buyer or sourcing	Looked up parts, prices, stock, and purchase links	Placing orders. Supplier contact. Substitutes when stock runs out	Less research time per order
Test engineer	Wrote the automated tests and 14 bench and field procedures	Running the procedures on hardware and judging the results	Less time writing procedures. Test execution is unchanged
Senior engineer or checker	Nothing. No one has reviewed or signed this package	Review, sign-off, and accountability for the design	More review work, not less, because drafts arrive faster
Technician or builder	Nothing. No physical work was done	All assembly, measurement, and testing	Unchanged

Two roles deserve a direct answer.

- **Drafters.** A company could reduce drafting hours for simple parts now. It should not remove the checking function.
- **Entry-level engineers.** A company could reduce routine desk tasks now. Eliminating the entry level removes the people who become the next reviewers and approvers.

Three facts from this demo should temper any headcount decision.

- **The design was simple.** It has one flat plate, two foam parts, stock components, and no custom electronics. The demo says nothing about complex 3D models or production drawings released to a shop.
- **Claude made mistakes.** The record shows 12 defects found and fixed during the work, including the two design faults. Claude found these itself. A company still needs people who can find the errors Claude does not catch.
- **Review became the bottleneck.** The drafts arrived within two days and still wait for a qualified person to check them.

Where the law or a customer requires a licensed engineer's approval, that requirement does not change.

The sound next step is a measured pilot. Run one real project with Claude drafting and staff reviewing, track hours and errors by role, and set staffing from that data.

Decisions and next steps

Leadership has one decision to make now: whether to fund the first physical build and bench test. After approval the work runs in this order, and each step has written pass criteria.

1. **Review and sign the drawings.** A qualified person checks the four sheets and completes the DRAWN, CHECKED, and APPROVED blocks.
2. **Order the parts.** Budget about $209 for one node and about $172 for test equipment. Recheck prices and stock first.
3. **Load and verify the firmware** on the array, following the firmware setup procedure.
4. **Prove audio capture.** Bench tests HB-1 and HB-2 confirm six-channel capture on Windows and the microphone order. This is the first go or no-go point. If Windows cannot provide six channels, the node moves to Linux.
5. **Build the mount** from the build guide and the drawings, and record every difference from the drawings.

6. **Calibrate the bearing** with a loudspeaker (HB-5), then run the remaining bench tests.
7. **Run the field tests:** background recording, wind, false alerts, controlled drone flights, and a two-node trial.

Thirteen engineering decisions remain open, such as the host computer, wind protection, and whether to train a classifier. The limitations document names the test that settles each one.

Before any field work, check the aviation rules for test flights and the audio recording laws for the site. This note is not legal advice.

Where to find everything

Start with the demo and the drawing set. File paths are inside the project folder `Passive_Acoustic_UAS_Sensor`.

Item	Location
Interactive demo, web copy	Passive Acoustic Sensor Demo. It opens only for people it is shared with
Interactive demo, file	`demo/index.html`. Opens in any browser, with no installation
Engineering drawing set	`hardware/drawings/SLM-PAS-DRAWING-SET.pdf`
Bill of materials	`hardware/BOM.md` and `hardware/bom.csv`
Hardware build guide	`hardware/HARDWARE_BUILD.md`
Firmware setup procedure	`firmware/PROVISIONING.md`
Requirements	`docs/REQUIREMENTS.md`
Bench and field test procedures	`docs/TEST_PROCEDURES.md`
Verification report	`docs/VERIFICATION_REPORT.md`
Limitations and open decisions	`docs/LIMITATIONS.md`
Operator instructions	`docs/OPERATING.md`
Release package	`dist/Passive_Acoustic_UAS_Sensor_v0.2.2.zip`

The release archive’s SHA-256 checksum is `3be522b999024167cc9124669a9a4e4af0812b8da61c34437dbcc07dad070985`.